

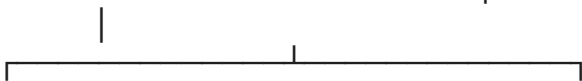
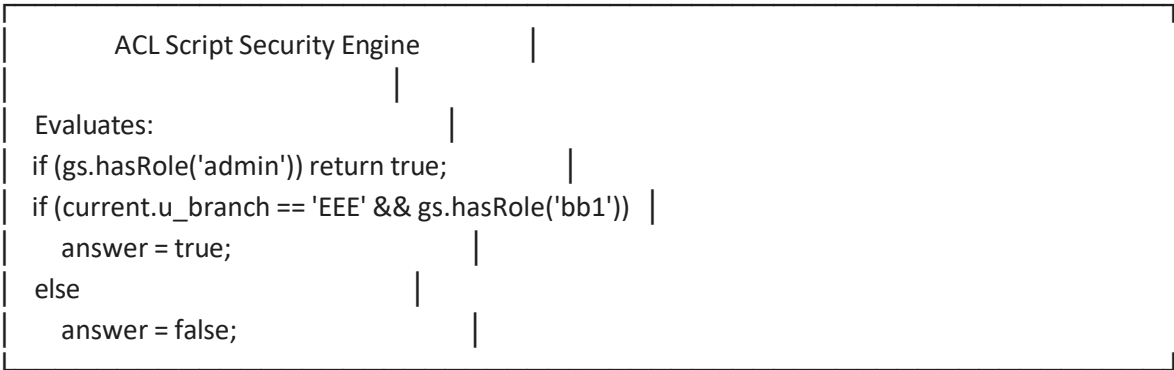
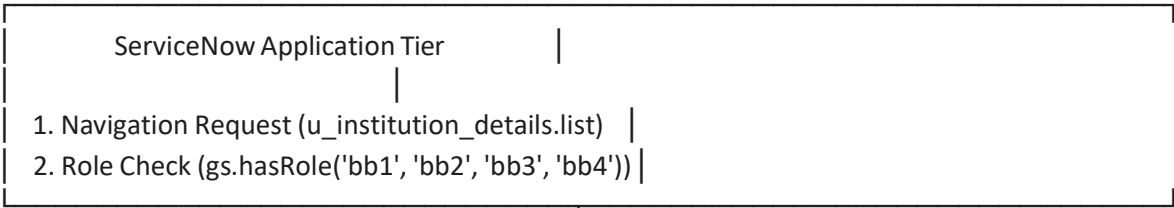
6. Prototype & Architecture Canvas

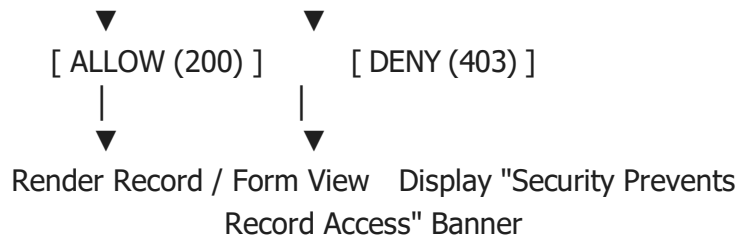
Prototyping Phase: Technical Architecture & System Workfilow

Header	Information
Date	02 October 2026
Team ID	
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	6 Marks

Technical Architecture Overview

[Client Browser / User]





Detailed Data & Security Schema

1. Custom Table Specification

- **Table Name:** Institution Details (u_institution_details)
- **Fields:**
 - Institution Name (u_name) — String
 - Branch (u_branch) — Choice List (EEE, CSE, ECE, MECH)
 - Code (u_code) — String
 - Active (u_active) — True/False

2. Scripted Access Control Rules

A. Read ACL (u_institution_details - Operation: read)

- **Requires Role:** bb1
- **Advanced Script:**

```
(function executeRule(current, previous /*null when async*/) {  
  // Grant access to administrators  
  ifi (gs.hasRole('admin')) {  
    answer = true;  
    return;  
  }  
  
  // Restrict read access based on field value 'EEE'  
  ifi (current.u_branch == 'EEE') {  
    answer = true;  
  } else {  
    answer = false;  
  }  
})(current, previous);
```

B. Create ACL (u_institution_details - Operation: create)

- **Requires Role:** bb2
- **Condition:** Role check enforced (gs.hasRole('bb2') || gs.hasRole('admin')).

C. Write ACL (u_institution_details - Operation: write)

- **Requires Role:** bb3
- **Condition:** Role check enforced (gs.hasRole('bb3') || gs.hasRole('admin')).

D. Delete ACL (u_institution_details - Operation: delete)

- **Requires Role:** bb4
- **Condition:** Role check enforced (gs.hasRole('bb4') || gs.hasRole('admin')).

Implementation & Testing Verifiication Matrix

Test Case	User Role	Record Branch Field	Expected Outcome	Actual Result	Status
TC-01	bb1	EEE	View Record Granted	Record Visible	PASS
TC-02	bb1	CSE	View Record Denied	Record Hidden	PASS
TC-03	bb2	Any	Insert Record Permitted	New Form Submission Allowed	PASS
TC-04	bb3	EEE	Update Record Permitted	Fields Editable	PASS
TC-05	bb4	EEE	Delete Record Permitted	Record Deleted	PASS
TC-06	No Role	Any	Access Denied	Security Error Displayed	PASS